

Shixin Song

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EDUCATION

Massachusetts Institute of Technology

Ph.D. Student in Computer Science

• Advisor: Prof. Mengjia Yan

2022 - Present

Cambridge, MA, USA

Massachusetts Institute of Technology

M.S. in Computer Science

• Advisor: Prof. Mengjia Yan

2022 - 2025

Cambridge, MA, USA

University of Michigan

B.S.E. in Computer Science

2020 - 2022

Ann Arbor, MI, USA

Shanghai Jiao Tong University

B.E. in Electrical and Computer Engineering

2018 - 2022

Shanghai, China

PEER-REVIEWED CONFERENCE PUBLICATION

SCHERI: Provably Secure Speculation Under the Constant-Time Policy for CHERI

2026

The ACM Conference on Computer and Communications Security (CCS) 2026

Shixin Song, Davide Davoli, Elias Storme, Marton Bognar, Dominique Devriese, Frank Piessens, Tamara Rezk

Securing Cryptographic Software via Typed Assembly Language

2025

The ACM Conference on Computer and Communications Security (CCS) 2025

Shixin Song*, Tingzhen Dong*, Kosi Nwabueze, Julian Zanders, Andres Erbsen, Adam Chlipala, Mengjia Yan

Oreo: Protecting ASLR Against Microarchitectural Attacks

2025

The Network and Distributed System Security (NDSS) Symposium 2025

Shixin Song, Joseph Zhang, Mengjia Yan

Thermometer: Profile-Guided BTB Replacement for Data Center Applications

2022

International Symposium on Computer Architecture (ISCA) 2022

Shixin Song, Tanvir Ahmed Khan, Sara Mahdizadeh Shahri, Akshitha Sriraman, Niranjan K Soundararajan, Sreenivas Subramoney, Daniel A. Jiménez, Heiner Litz, Baris Kasikci

RESEARCH EXPERIENCE

Verifying Out-of-Order Processors

2025 - Present

Developing the first machine-checked proof of out-of-order processors, which covers both functional correctness and timing side-channel security, in the style of cryptographic constant time

Secure Speculation for CHERI

2026

Developed a formal framework for jointly reasoning about capability safety, speculative execution, and information-flow security

Presented a new processor design within this framework and formally proved that it provides end-to-end secure speculation guarantees under the constant-time policy

Paper accepted at CCS'26

Securing Cryptographic Software

2023 - 2025

Developed a novel transformation technique for cryptographic software by inferring types and rewriting at the assembly level

Enabled a hardware mitigation for transient execution attacks with negligible overhead, achieving over $2\times$ speedup compared to prior solutions

Paper accepted at CCS'25

Software-Hardware Co-Design for Secure Systems

2022 - 2024

Developed a novel software-hardware co-designed mitigation that strengthens memory safety against microarchitectural attacks with negligible timing overhead and area cost

Paper accepted at NDSS'25

Improving Front-End Processor Performance for Data Center Applications	2021
Developed a novel BTB replacement policy which outperforms the state-of-the-art by 5.6x and achieves near-ideal front-end processor performance for data center applications	
<i>Paper accepted at ISCA '22</i>	
<i>Won the first place award at MICRO'21 ACM Student Research Competition</i>	
Enabling Early Hardware Development for Futuristic Data Center Applications	2021
Characterized widely-used data center applications (e.g., MySQL, MongoDB, FFmpeg, Nginx) and predicted how these applications might evolve in the future to enable suitable hardware optimization development early on	

AWARDS AND HONORS

EECS MathWorks Fellowship	2025
<i>Massachusetts Institute of Technology</i>	
Ho-Ching and Han-Ching Fund Award	2025
<i>Massachusetts Institute of Technology</i>	
Presidential Graduate Fellowship Award	2022
<i>Massachusetts Institute of Technology</i>	
CRA Outstanding Undergraduate Researcher Award Honorable Mention	2022
<i>Computing Research Association</i>	
ACM Student Research Competition First Place Winner	2021
<i>MICRO'21</i>	
Roger King Scholarship	2021
<i>University of Michigan</i>	<i>Ann Arbor, MI, USA</i>
Dean's List	2020, 2021, 2022
<i>University of Michigan</i>	<i>Ann Arbor, MI, USA</i>
SJTU Undergraduate Excellence Scholarship	2019, 2020
<i>Shanghai Jiao Tong University</i>	<i>Shanghai, China</i>
Fuda Scholarship	2019
<i>Shanghai Jiao Tong University</i>	<i>Shanghai, China</i>

EXPERIENCE

Teaching Assistant	2025
<i>Massachusetts Institute of Technology</i>	<i>Cambridge, MA, USA</i>
Secure Hardware Design (6.5950)	
Undergraduate Research Assistant	2021
<i>University of Michigan</i>	<i>Ann Arbor, MI, USA</i>
Advisor: Prof. Baris Kasikci	
Grader	2021 - 2022
<i>University of Michigan</i>	<i>Ann Arbor, MI, USA</i>
Teaching Assistant	2019 - 2021
<i>Shanghai Jiao Tong University</i>	<i>Shanghai, China</i>

AREAS & SKILLS

Areas: computer architecture and security, formal verification, type system
Skills: C/C++, Python, OCaml, Rocq, SystemVerilog, gem5, ChampSim, Z3

COURSE WORK

Massachusetts Institute of Technology

- * 6.5900: Computer System Architecture, 6.5120: Formal Reasoning about Program, 6.5620: Cryptography & Cryptanalysis, 6.8300: Advances in Computer Vision

University of Michigan

- * EECS 470: Computer Architecture, EECS 482: Introduction to Operating System, EECS 475: Introduction to Cryptography

Shanghai Jiao Tong University

- * VE 280: Programming and Elementary Data Structures